

**Major Project Stage-1 Zeroth Review Presentation
on**

**Predictive Thermal Battery Management System
for Li-ion Battery Packs**

Presented by

Ms. Muskan Sulathana
Roll No. : 23WH1A0208

Guide

Dr. M. Rupesh
Associate Professor
EEE Department
BVRIT HYDERABAD College of Engineering
for Women, Hyderabad

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Abstract

- Lithium-ion cells are used in phones, laptops and electric vehicles, and they can catch fire if they are allowed to get too hot.
- Most small protection boards switch the pack off only after the temperature crosses a fixed value, usually about 60 °C.
- By the time a cell reaches that value it is often already heating itself, and that heat then causes still more heating.
- This project builds a battery management system for a four-cell lithium-ion pack that watches how fast the temperature is rising, not only how high it has reached.
- An ESP32 reads the cell voltages, the pack current and four cell temperatures, switches the pack off through MOSFETs when the rise is too fast, and sends every reading over Wi-Fi to a simple web dashboard.

Problem Statement

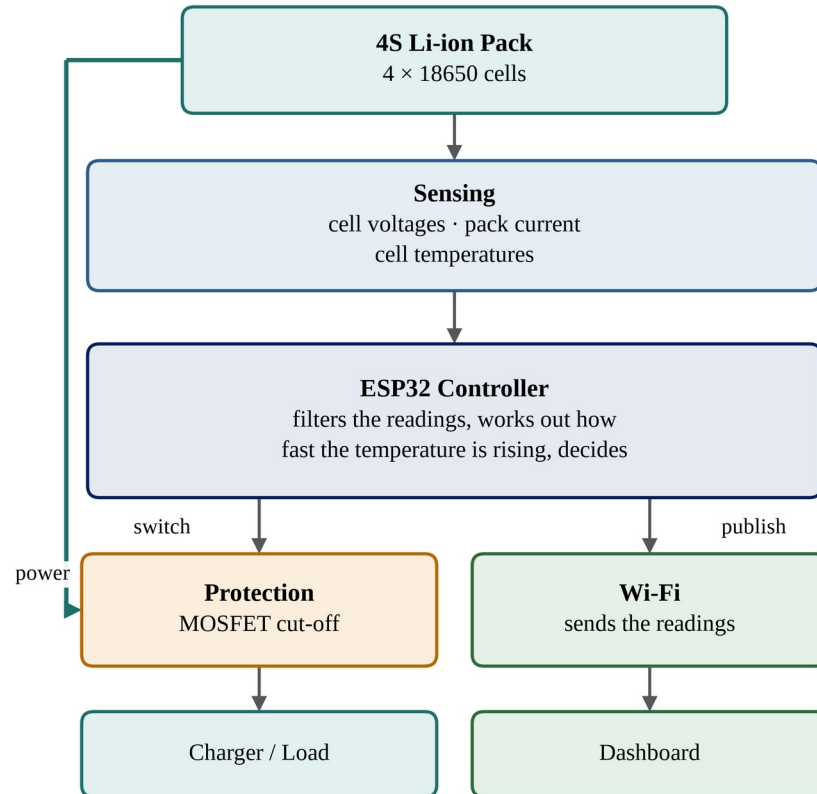
- A fixed temperature limit only acts once the cell is already hot, and it gives no warning before that point.
- A cell that is heating quickly at 35 °C is already faulty, but it passes a 60 °C test without any complaint.
- The sensor sits on the outside of the cell, so the inside is always hotter than the reading suggests.

Objective

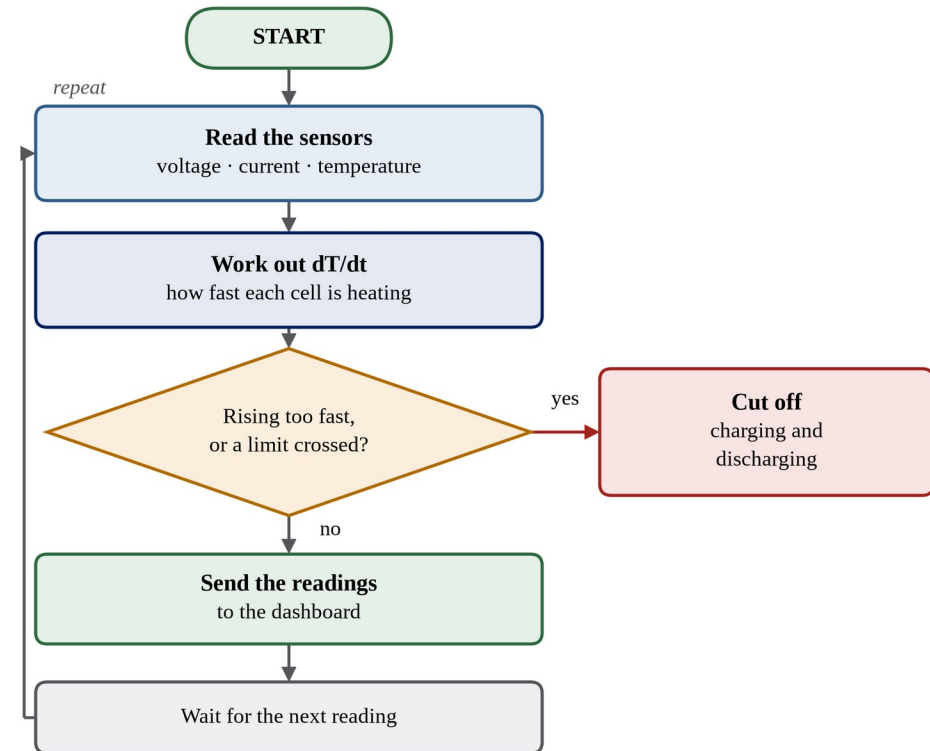
- To design and build a battery management system for a 4S lithium-ion pack that measures cell voltage, pack current and cell temperature, cuts off charging and discharging based on how fast the temperature is rising, and publishes every reading live over Wi-Fi.

Implementation Outline

Block diagram



Firmware flowchart



Components

Component	How many	What it does
18650 lithium-ion cells	4	The battery pack the system looks after
ESP32 board	1	Reads the sensors and decides when to cut off
NTC thermistors	4	Measure the temperature of each cell
Resistor dividers	4	Bring each cell voltage down to a level the ESP32 can read
Current sensor with shunt	1	Measures the current going into and out of the pack
Analogue multiplexer	1	Lets one input pin read all four thermistors
MOSFETs	2	The switches that stop charging and discharging
Bleed resistors	4	Drain a little charge from a cell that is fuller than the rest
Buck converter	1	Makes the 3.3 V supply for the ESP32 from the pack
Fuse	1	The last protection if the electronics fail

Implementation and Tools

This project has a hardware part and a software part. Both are built here.

Side	Tool	What it is used for
Hardware	FreeCAD	Modelling the enclosure
Hardware	3D printer, PLA+	Printing the enclosure
Hardware	Soldering station	Building the board, perfboard first, then a soldered PCB
Hardware	Multimeter, bench supply	Measuring and testing the circuit
Software	ESP-IDF	The framework the firmware is built on
Software	C++	The language the firmware is written in
Software	FreeRTOS	Running the sensor, safety and telemetry tasks
Software	HTML, CSS, JavaScript	The dashboard page in the browser
Software	Git and GitHub	Keeping the code and the documents

Literature Survey

References	Summary of the work
[1]	How thermal runaway starts inside a lithium-ion cell, and the temperature at which each stage begins.
[2]	A review of the ways thermal runaway can be predicted early, and what each method needs in order to work.
[3]	A warning method that combines data clustering with a heat model, and warns 25 minutes before thermal runaway.
[4]	A review of what a battery management system does, and which problems in it are still open.
[5]	The Indian safety standard for sealed lithium cells, and the abuse tests it lays down.

References

- [1] X. Feng, M. Ouyang, X. Liu, L. Lu, Y. Xia and X. He, “Thermal runaway mechanism of lithium ion battery for electric vehicles: A review,” *Energy Storage Materials*, vol. 10, pp. 246–267, 2018.
- [2] X. Zhang, S. Chen, J. Zhu and Y. Gao, “A critical review of thermal runaway prediction and early-warning methods for lithium-ion batteries,” *Energy Material Advances*, vol. 4, art. 0008, 2023.
- [3] Q. Chen, Y. He, N. Fang and G. Yu, “A combined data-driven and model-based algorithm for accurate battery thermal runaway warning,” *Sensors*, vol. 24, no. 15, art. 4964, 2024.
- [4] A. K. M. A. Habib, M. K. Hasan, G. F. Issa, D. Singh, S. Islam and T. M. Ghazal, “Lithium-ion battery management system for electric vehicles: Constraints, challenges, and recommendations,” *Batteries*, vol. 9, no. 3, art. 152, 2023.
- [5] IS 16046 (Part 2) : 2018 / IEC 62133-2 : 2017, Secondary cells and batteries containing alkaline or other non-acid electrolytes — Part 2: Lithium systems.

Thank You

Questions are welcome